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The Collaborative UK ECMO Trial: Follow-up to 1 Year of Age

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ABSTRACT. *Objective.* To evaluate the clinical effectiveness of neonatal extracorporeal membrane oxygenation (ECMO), in terms of mortality and morbidity, in the treatment of cardiorespiratory failure in term infants.

Methods. The criteria for trial entry were: an oxygenation index of >40 or arterial partial pressure of carbon dioxide (P_{aCO_2}) >12 kPa for at least 3 hours; gestational age at birth of 35 completed weeks or more; a birth weight of 2 kg or more; <10 days high-pressure ventilation; an age of <28 days; and no contraindication to ECMO such as previous cardiac arrest or intraventricular hemorrhage. Eligible infants were randomized either to be transferred to one of five ECMO centers in the United Kingdom or to continue conventional treatment. The principal outcome was death or severe disability at the age of 1 year. Severe disability was defined as an overall developmental quotient of <50 using the Griffiths Mental Development Scales, or blindness or a level of function so as to make assessment using the Griffiths Scales impossible.

Families of surviving children were contacted at regular intervals during the first year and at the age of 1, and an assessment of the child was performed by one of three developmental pediatricians. This included a neurologic examination, assessment of hearing and vision, developmental level, general health, and health service use.

Results. Of 185 infants recruited into the trial, 93 infants were in the ECMO arm and 92 were allocated conventional treatment. The groups were comparable at trial entry. Thirty of 93 (32%) ECMO infants died before the age of 1 year and 54 of 92 (59%) of the infants in the conventional group died. Two infants were lost to follow-up, 1 from each arm of the trial.

Of the remaining 99 survivors, at the age of 1 year, 2 infants (1 in each arm) were still in the hospital, and 5 (3 in the ECMO arm and 2 conventional) still required supplementary oxygen. Fifteen infants had tone changes in the limbs, 10/62 (16%) in the ECMO arm and 5/37 (13.5%) in the conventional arm. These signs were more common on the left side in both groups. One infant (in the ECMO arm) had bilateral sensorineural deafness and 1 infant (also in the ECMO arm) had low vision.

Overall, 2 infants were severely disabled (1 ECMO and 1 conventional), 16 others also had evidence of functional loss (12 vs 4), and 8 had impairment without functional loss (4 vs 5). There was a trend toward proportionately greater respiratory morbidity in the conventional group. Neurologic morbidity was more common in the ECMO group, reflecting the larger number of survivors.

The lower rate of adverse primary outcome (death or severe disability at 1 year) was found among infants allocated ECMO in all the predefined stratified analyses. Disease severity at trial entry and type of referral center did not appear to alter the effects of ECMO. Only 4 of 18 infants with congenital diaphragmatic hernia survived and at age 1 year only 1 of the 4 survivors was considered normal.

Conclusion. These results are in accord with the earlier preliminary findings that a policy of ECMO support reduces the risk of death without a concomitant rise in severe disability. However, 1 in 4 survivors had evidence of impairment with or without disability. Further follow-up is planned at the age of 4 and 7 years. *Pediatrics* 1998;101(4). URL: <http://www.pediatrics.org/cgi/content/full/101/4/e1>; extracorporeal membrane oxygenation, mortality, morbidity, neonate.

ABBREVIATIONS. ECMO, extracorporeal membrane oxygenation; RR, relative risk; CI, confidence interval; CT, conventional treatment; DQ, developmental quotient.

Preliminary results of the UK collaborative randomized trial of neonatal extracorporeal membrane oxygenation (ECMO) were published in July 1996.¹ The trial was mounted in response to considerable uncertainty about the clinical and cost effectiveness of neonatal ECMO in the treatment of mature newborn infants with potentially reversible severe cardiorespiratory failure. Recruitment was started in 1993 and infants were randomized to one of two policies: 1) to be transferred to one of five ECMO centers in the UK for consideration of ECMO support, or 2) to continue conventional treatment in a neonatal intensive care unit.

The principal endpoint was defined as death or severe disability at the age of 1 year because there was a concern that ECMO might reduce mortality at the expense of severe disability among survivors. This might occur for three reasons. First, infants who are considered for ECMO have often had episodes of severe hypoxia and cardiac compromise as a result of their cardiorespiratory problems. Second, ECMO involves placing cannulae in the internal carotid artery and/or internal jugular vein, with possible impairment of cerebral perfusion. Third, ECMO requires that the infant is heparinized with the associated risk of hemorrhage.

Recruitment to the trial was stopped in November 1995 after 185 infants had been recruited. It had become clear that sufficient data were available from the 1-year follow-up to be sure that any differences in severe disability would not be large enough to counterbalance the differences in death rates between the

From the National Perinatal Epidemiology Unit, Radcliffe Infirmary, Oxford, England.

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Address correspondence to: Ann Johnson, MD, FRCP, National Perinatal Epidemiology Unit, Radcliffe Infirmary, Oxford, OX2 6HE England.

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two groups: the relative risk (RR) of death in the ECMO group compared with the conventionally treated group was 0.55 (95% confidence interval [CI] 0.39 to 0.77). Based on information about those infants recruited at least a year earlier, that is, in 1993 and 1994, the RR for the principal outcome (death or severe disability) was 0.54 (95% CI, 0.36–0.80). This conclusion was provisional as, at that time, the 1-year outcome was not ascertainable for all infants recruited. We report here the full results of the UK ECMO trial with details of later deaths and the status of all survivors at the age of 1 year.

METHODS

The full details of the trial organization and treatment schedules have been reported previously.¹ In summary, the criteria for trial entry were: severe respiratory failure (defined as an oxygenation index of >40 or arterial partial pressure of carbon dioxide [P_{aCO_2}] >12 kPa for at least 3 hours); gestational age at birth of 35 completed weeks or more and a birth weight of 2 kg or more; <10 days of high-pressure ventilation; an age of <28 days; and no contraindication to ECMO support such as previous cardiac arrest or severe intraventricular hemorrhage. When parental consent was obtained, infants were randomly allocated ECMO or conventional treatment using a computerized minimization algorithm to ensure balance on the following prognostic variables: primary diagnosis; disease severity; referral center; and the ECMO center where a cot had been reserved.

The majority of infants allocated to ECMO were transported by road ambulance, fixed wing aircraft, or helicopter to an ECMO center, usually by a team from that center. ECMO was started as soon as possible after arrival unless a major deterioration or improvement in the infant's condition or detection of a cardiac anomaly made ECMO support inappropriate. Vascular cannulation was either veno-arterial via the right internal carotid artery and the right internal jugular vein, or veno-veno (right internal jugular). Subsequent treatment followed an agreed protocol. Cannulated vessels were repaired where possible when ECMO support was discontinued. Infants who survived were usually transferred back to the hospital from which they were originally referred after completion of the ECMO run.

Infants allocated to conventional treatment (CT) were treated in the referring hospital. Treatment guidelines recommended liberal use of oxygen, adequate ventilation, correction of acidosis, maintenance of adequate blood pressure, paralysis, and use of available pulmonary vasodilators, including nitric oxide. Use of high-frequency ventilation and surfactant was left to the clinicians' discretion.

Permission to carry out a postmortem examination was requested from parents of infants who died. When possible, a more detailed examination of lung tissue, cardiac muscle, and brain was performed, and these findings will be reported separately.

Follow-up Procedure

The parents of all infants who survived to discharge were contacted by the trial coordinator just before their infant's discharge to explain the follow-up procedure in the first year. Families were contacted again by telephone when the infant reached the ages of 4 months, 8 months, and 11 months. Before each phone call, the child's health visitor (a community-based nurse with particular responsibility for preschool children) was asked if such a contact was appropriate. Families without a telephone were given a phone card and reminded by letter to contact the coordinating office at the appropriate times.

At each telephone contact, the parent was asked about the infant's overall health and their use of health services since the previous contact. At 11 months, arrangements were made for a pediatrician to visit the home to carry out a neurodevelopmental assessment of the child. This assessment consisted of a neurologic examination, developmental assessment using the Griffiths Mental Development Scale,² measurement of weight and head circumference, assessment of vision and hearing, and questions about seizures, respiratory symptoms, readmission to hospital, and health service use including referral to specialists. Each assess-

ment was performed by one of three developmental pediatricians following standardized procedures agreed on beforehand. Usually they traveled to the homes of the infants although if the infant had not been discharged home by the age of 1 year, the assessment was performed in the hospital. The pediatricians were not aware of the trial allocation or the infant's neonatal course. Before the assessment, parents were given a high-necked tunic for the infant to wear during the examination so that any scars on the neck at the site of cannulation were hidden from the examiner. After the assessment, the results were sent to the infant's general practitioner, the health visitor, and local pediatrician, with a copy to the parents if requested. If the infant had been seen for a suspected vision or hearing problem in a specialist clinic, the results of these examinations were requested.

After the neurodevelopmental assessment and if parents agreed, arrangements were made for the infant to have respiratory function tests in one of two centers. These findings are reported elsewhere.³

Finally, to facilitate further follow-up, all infants were flagged on the NHS Central Register (which will allow their family doctor to be identified later), and parents were asked to let the trial office know about any change of address.

Economic Evaluation

The economic evaluation, carried out alongside the trial, was an integral feature of the trial design. Its objective was to assess the cost effectiveness of ECMO compared with conventional treatment; that is the additional cost, if any, of achieving any observed benefit. The methods used have already been published.⁴ The full results of the economic evaluation will be reported elsewhere.

Definitions of Outcomes

The principal endpoint was death before, or severe disability at, the age of 1 year. Severe disability was defined as an overall developmental quotient (DQ) of <50 on the Griffiths Mental Developmental Scales, or blindness or a level of function so as to make assessment using the Griffiths Scales impossible. The remaining infants were categorized into three groups: a) Impairment with disability not classified as severe—this included infants with abnormal motor function, for example those with asymmetry of limb use or motor DQ <70 , infants with global delay (DQ 50–69), those on regular anticonvulsants, or those requiring tube-feeding or oxygen at the age of 1 year; b) Impairment without disability—this included infants with abnormalities of tone or posture but no evidence of functional loss at this age; and c) Normal—no evidence of impairment or disability.

Statistical Analysis

The analyses are based on the groups as randomized, with secondary analysis stratified by the factors used for minimization at trial entry. Differences between groups are presented as RRs with 95% CIs. Percentages for the outcomes assessed at 1 year in the randomized groups are based either on those seen at 1 year, or on all those randomized depending on which was most appropriate.

RESULTS

A total of 185 infants were recruited into the trial, 93 were allocated to ECMO, and 92 were allocated to conventional treatment. As shown in Fig 1, 82 infants died before discharge from hospital, 2 more died between discharge and the age of 1 year, and 2 infants were lost to follow-up. A total of 99 infants were assessed at the age of 1 year.

Description and Comparability of Trial Groups

Overall 19% (35/185) of trial infants had a congenital diaphragmatic hernia, 37% (69/185) had meconium aspiration, two fifths had severe lung disease as indicated by an oxygenation index of 60 or more, and 78% of infants were recruited before the age of

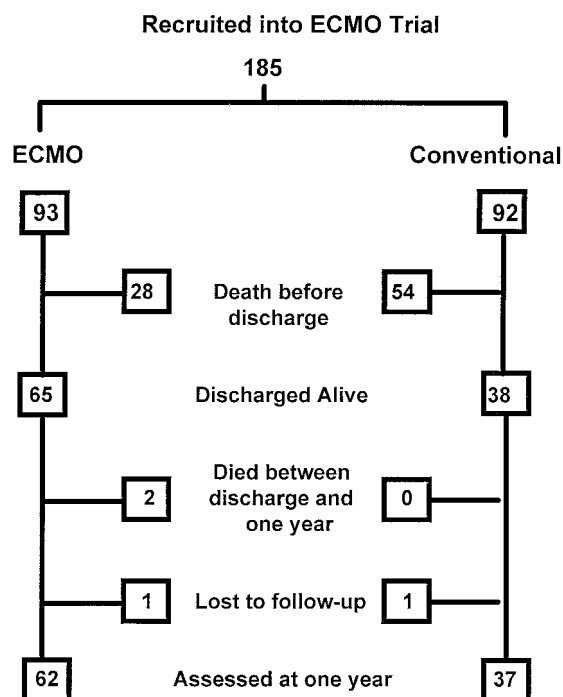


Figure. Numbers of infants in ECMO trial.

48 hours. The trial groups as randomized were well-matched in respect of known prognostic variables (Table 1).

The characteristics at the time of trial entry of those infants who were assessed at the age of 1 year, also shown in Table 1, reflected differences in the chances of survival associated with the descriptive variables. In particular, only 4 of the 35 infants with congenital diaphragmatic hernia survived to the age of 1 year. Differences between trial groups at this time were

caused by the increased survival associated with ECMO support.

Treatment Between Trial Entry and Discharge Home

Of 93 infants allocated ECMO, 78 actually received ECMO support. The reasons for not giving ECMO are shown in Table 2. Five infants died before transfer from the referring hospital to an ECMO center; none died during transportation. One infant allocated to conventional care had ECMO. The larger number of early deaths in the conventional treatment group resulted in a higher median number of days on a ventilator and days in the hospital in the all randomized ECMO group. Among survivors, however, the median number of days on a ventilator in high oxygen, and median duration of hospital stay were lower in the ECMO arm.

Deaths

Of eighty-four infants who were known to have died, 30 were in the ECMO arm of the trial and 54 in the conventional arm (RR, 0.55; 95% CI, 0.39–0.77; $P = .0005$). Forty-nine (58%) died in the first 7 days after birth and 82 (98%) before hospital discharge. The two deaths after discharge were among infants allocated to ECMO. Information on cause of death from postmortem examination was available for 22 of the 30 (73%) infants who died in the ECMO arm and 34 of the 54 (63%) infants in the conventional treatment arm. There were fewer deaths in the ECMO group and this pattern was seen in most subgroups. It was most marked among infants with congenital diaphragmatic hernia, meconium aspiration (particularly those with severe lung disease but no clear evidence of other organ damage), and among those with idiopathic respiratory distress syndrome (Table 3).

TABLE 1. Comparability of Trial Groups

	All Randomized		Infants Assessed at 1 Year	
	ECMO (93)	CT (92)	ECMO (62)	CT (37)
Primary diagnosis				
Congenital diaphragmatic hernia	18 (19)	17 (18)	4 (6)	0
Idiopathic persistent fetal circulation	15 (16)	16 (17)	13 (21)	6 (16)
Persistent pulmonary hypertension attributable to:				
Meconium aspiration	32 (34)	37 (40)	25 (40)	20 (54)
Sepsis	14 (15)	5 (5)	9 (15)	1 (3)
Idiopathic respiratory distress syndrome	8 (9)	12 (13)	7 (11)	7 (20)
Other	3 (3)	4 (4)	3 (5)	2 (5)
Other	3 (3)	1 (1)	1 (2)	1 (3)
Entry criteria				
OI 40–59	55 (59)	55 (60)	41 (66)	25 (68)
OI ≥ 60	37 (40)	37 (40)	20 (32)	12 (32)
Paco ₂ >12 kPa for >3 h	1 (1)	0	1 (2)	0
Age at entry				
Less than 1 day	37 (40)	42 (46)	23 (37)	18 (49)
1 day	31 (33)	20 (22)	17 (27)	7 (19)
2–27 days	25 (27)	30 (33)	22 (35)	12 (32)
Sex				
Male	54 (58)	58 (63)	39 (63)	23 (62)
Female	39 (42)	34 (37)	23 (37)	14 (38)
Gestational age in completed weeks				
Median [IQR]	39 [37–40]	40 [38–41]	39 [37–40]	40 [38–41]
Birth weight (g)				
Mean [SD]	3261 [516]	3346 [593]	3380 [500]	3568 [594]

Number (%) unless otherwise indicated.

TABLE 2. Management After Trial Entry

	All Randomized		Infants Assessed at 1 Year	
	ECMO (93)	CT (92)	ECMO (62)	CT (37)
ECMO support				
Given	78 (84)	1 (1)	57 (92)	0
Not given because:				
Too well	3 (3)	—	3 (5)	—
Too ill	2 (2)	—	0	—
Died before transfer	5 (5)	—	0	—
Congenital heart disease	4 (4)	—	0	—
Failed cannulation	1 (1)	—	0	—
Days on ventilator				
Median {IQR}	2 {0.5,4}	0 {0,5}	2 {1.5,3.6}	5.5 {3,10}
Days in >90% oxygen				
Median {IQR}	0.5 {,5,1.0}	2 {1,5}	0.5 {0.5,1}	3 {2,6.5}
Days in supplementary oxygen				
Median {IQR}	3 {0,12.5}	0 {0,5}	6.3 {2.4,14.9}	8 {2.5,21}
Days in hospital				
Median {IQR}	24 {15.5–37}	8.5 {1,29}	24 {19,37}	30 {18,43.5}

Number (%) unless otherwise indicated.

TABLE 3. Infants Who Died Before the Age of 1 Year

	ECMO (93)	CT (92)
Death before age of 1 year, number (%)	30 (32)	54 (59)
Age at death		
<7 days	9	40
7–28 days	11	11
29–60 days	3	2
>60 days	7	1
Cause of death		
Congenital diaphragmatic hernia		
Preoperative	4	13
Postoperative	7	4
Pulmonary hypoplasia/dysplasia	1	4
Congenital heart disease	5	1
Sepsis		
Group B streptococcal septicemia	3	2
Streptococcal pneumonia	0	1
Pseudomonas	1	1
Herpes simplex	0	1
Chickenpox pneumonitis	0	1
Candida meningitis	1	0
Meconium aspiration		
With evidence of cerebral or other organ hypoxic damage	4	3
No clear evidence of cerebral or other organ hypoxic damage	0	13
Idiopathic respiratory distress syndrome	0	6
Pulmonary hemorrhage	0	2
Intrauterine hypoxia	0	1
Other		
Absent surfactant protein B deficiency	0	1
Congenital pulmonary lymphangiectasis	2	0
Myotonic dystrophy	1	0
Cardiac tamponade	1	0

Status of All Survivors at Age 1 Year

There were 101 survivors: 63 in the ECMO group and 38 in the conventional group. Two infants were lost to follow-up. The parents of 1 infant allocated to ECMO refused follow-up but the infant was considered normal when seen in a pediatric clinic at the age of 16 months. The second, allocated conventional treatment, returned to Bangladesh at the age of 8 months. He was reported as well by his father when telephone contact was made at the age of 4 months. At the age of 1 year, 2 infants were still in the hospital, 1 in each trial group. Both were fully assessed.

A higher proportion of survivors in the conventional management arm showed signs of respiratory

morbidity (Table 4), although only the use of bronchodilators reached statistical significance ($P = .02$). Of the 11 infants who were discharged from hospital on supplementary oxygen, 5 (3 in the ECMO arm) still required oxygen for at least part of each 24 hours at 1 year of age. Both infants still in the hospital at this age (1 in each trial group) were being partially or totally fed by nasogastric tube, as well as 2 more at home in the ECMO arm of the trial. The weight of 12 of the children (2 with congenital diaphragmatic hernia) was below the third centile. One infant in the ECMO arm had low vision associated with cerebral palsy, although there was also evidence of partial albinism. Another infant, also in the ECMO arm, had bilateral sensorineural deafness with hearing aids.

TABLE 4. Status of Survivors Assessed at 1 Year

	ECMO (62)	Conventional (37)
Respiratory		
Cough at night	16 (26)	14 (38)
Wheeze	7 (11)	8 (22)
Use regular bronchodilators	5 (8)	10 (27)
Additional oxygen at home during first year	5 (8)	6 (16)
Still on oxygen at 1 year	3 (5)	2 (5)
Chest infection since discharge home	39 (63)	29 (78)
Number of readmissions to hospital for respiratory reasons	31 —	26 —
Feeding		
Infants needing tube feeding	3 (5)	1 (3)
Growth		
Weight		
<3rd centile	9 (15)	3 (8)
Head circumference		
<3rd centile	4 (6)	1 (3)
>97th centile	5 (8)	2 (5)
Vision		
Considered to have normal vision	53 (84)	35 (96)
Squint	3 (5)	2 (5)
Unable to see 5-mm object	1 (2)	0
Uncertainty about vision	5 (8)	0
Hearing		
Appears normal	50 (81)	25 (68)
Has hearing aid	1 (2)	0
Uncertainty about hearing	11 (18)	12 (32)
Seizures		
On anticonvulsants	2 (3)	0
Neuromotor		
Tone change affecting		
4 limbs	4 (6)	2 (5)
3 limbs	0	1 (3)
2 limbs: contralateral	2 (3)	0
ipsilateral	3 (5)	1 (3)
1 limb	1 (2)	1 (3)
Signs predominantly on		
Right side	5 (8)	0
Left side	7 (11)	3 (8)
Abnormal axial tone	7 (11)	3 (8)
Abnormal movements present	2 (3)	0
Griffiths		
Motor DQ		
85 or more	52 (84)	34 (92)
70–84	5 (8)	0
50–69	3 (5)	1 (3)
<50	1 (2)	2 (5)
Unable to test	1 (2)	0
Griffiths		
Overall DQ		
85 or more	57 (92)	34 (92)
70–84	3 (5)	1 (3)
50–69	1 (2)	1 (3)
<50	0	1 (3)
Unable to test	1 (2)	0

Number (%) unless otherwise indicated.

There was uncertainty about hearing for a quarter of all infants seen. Two infants (both in the ECMO arm) were on regular anticonvulsants.

Fifteen infants had evidence of tone change in the limbs, 10/62 (16%) in the ECMO arm and 5/37 (13.5%) in the conventional treatment group. There were more infants with neurologic signs predominantly on the left side (10/99) than on the right (5/99), but the proportion of infants with left sided signs was no greater in the ECMO group than in the conventional treatment group.

One infant in the ECMO group was considered too disabled to test using the Griffiths Mental Development Scales. He had spastic quadraparesis, low vision, seizures, and developmental delay. One infant

in the conventional treatment group had global delay and an overall Griffiths DQ of 47. Two children had a DQ between 50 and 69. One (in the conventional arm) was still in the hospital and was oxygen-dependent with chronic lung disease and a gastrostomy. The second, in the ECMO arm, had hypopituitarism and lung hypoplasia; he was oxygen-dependent, required tube-feeding and was globally delayed. Overall, 92% of the surviving infants had an overall Griffiths DQ of 85 or more.

Overall Status at the Age of 1 Year

Using all the information available, all the infants were categorized as shown in Table 5. Details of the 17 infants in the ECMO arm and 10 infants in the

TABLE 5. Overall Status at 1 Year of Age—Number (%)

	ECMO (93)	CT (92)
Deaths	30 (32)	54 (59)
Lost to follow-up	1 (1)	1 (1)
Children with severe impairment and disability	1 (1)	1 (1)
Signs of impairment		
With disability not classified as severe	12 (13)	4 (4)
Without disability	4 (4)	5 (5)
No signs of impairment or disability	45 (48)	27 (29)
Number of known survivors with no signs of impairment or disability (% of survivors)	45/62 (73)	27/37 (73)

conventional treatment group who were not classified as normal at the age of 1 year are in the Appendix. The principal outcome was death or severe disability at the age of 1 year. The known rate of this adverse outcome in the ECMO arm was 31/93 (33%), and in 55/92 (60%) in the conventionally managed arm (RR, 0.56; 95% CI, 0.40–0.78; $P = .0005$).

The lower rate of adverse primary outcome was found among infants allocated ECMO in all the predefined stratified analyses. The benefit of ECMO is most clearly seen among those with respiratory failure other than that caused by diaphragmatic hernia. Only 4 children with diaphragmatic hernia survived (all in the ECMO group) and at the age of 1 year, 3 had evidence of impairment with disability, and only 1 infant was considered normal. Disease severity at trial entry and type of referral center did not appear to alter the effects of ECMO, although numbers are small (Table 6).

DISCUSSION

This is the only report of a randomized controlled trial of an ECMO policy that has published details of a longer-term follow-up. The full results are in accord with those we reported in 1996, and with a 2-year follow-up from another trial that has been reported in abstract only.⁵ A policy of ECMO support reduces the risk of mortality without a major concomitant rise in severe disability by about 45% (95% CI, 60%–22%), regardless of the severity of the infant's condition.

Just under half of all infants randomly allocated ECMO and two thirds of all survivors were consid-

ered normal at 1 year of age. Morbidity was most evident in the respiratory and neurologic systems. Further information about lung function will be available from the respiratory follow-up study, but the small number of infants who still needed oxygen at home at the age of 1 year clearly had overt evidence of severe persisting lung damage. The trend toward a higher proportion of surviving infants in the conventional treatment arm having a tendency to wheeze and more frequent respiratory infections might suggest that continued ventilation at high pressure with high concentrations of oxygen contribute to respiratory morbidity at the end of the first year. Whether these symptoms persist into childhood and their relationship with the results of later lung function tests will be known after the planned assessment at the age of 7 years.

In terms of neurologic function, one of the concerns was the possibility of long-term adverse effects of cannulation and subsequent ligation or repair of major vessels in the right side of the neck.⁶ There is conflicting evidence about the frequency of lateralized cerebral lesions among infants who have had ECMO.^{7–9} In this trial, the number of infants with left-sided signs in the ECMO group was not significantly larger than in the conventional arm, although numbers are small.

A further concern has been the risk of intracranial hemorrhage secondary to heparinization and circulatory disruption among infants who have ECMO. Infants with severe hemorrhagic or ischemic lesions on cranial ultrasound were ineligible for trial entry, but 38 infants (27 in the ECMO arm and 11 in the

TABLE 6. Outcome at Age 1 Year by Primary Diagnosis, Disease Severity, and Referral Center Type

	Babies Assessed at 1 Year							
	Severe Disability		Impairment				Normal	
	ECMO	CT	Functional Loss		No Functional Loss		ECMO	CT
			ECMO	CT	ECMO	CT		
All infants	1	1	12	4	4	5	45	27
Primary diagnosis								
Congenital diaphragmatic hernia	0	0	3	0	0	0	1	0
Idiopathic persistent fetal circulation	0	0	4	1	0	2	9	3
Meconium aspiration	1	0	3	1	1	2	20	17
Other	0	1	2	2	3	1	15	7
Disease severity								
Oxygenation index <60	0	0	8	3	2	4	32	18
Oxygenation index ≥60	1	1	4	1	2	1	13	9
Referral center type								
Teaching	1	1	10	2	0	3	29	17
Nonteaching	0	0	2	2	4	2	16	10

conventionally managed group) showed lesions on cranial ultrasound after trial entry. This difference may reflect the early death of a number of infants in the conventional arm, before there was an opportunity for further investigation. Almost half of the infants who developed ultrasound lesions after trial entry died, (11 of 27 of the ECMO infants and 5 of the 11 conventional treatment group). Six of the 16 ECMO survivors and 1 of the 6 conventional treatment survivors with ultrasound lesions had evidence of impairment at the age of 1 year.

Sensorineural hearing loss, especially high-frequency loss has been reported in association with both persistent pulmonary hypertension and after ECMO therapy.^{10,11} Many of the survivors in this trial had only a distraction test to identify hearing loss, and the degree of uncertainty about hearing status in nearly a quarter of the infants tested, probably reflects the limitation of this type of assessment. Audiometry is planned for all children at the age of 4 years.

A number of the children had tone changes and asymmetry, some with mild motor delay. Although many of these infants appear to be functionally normal at the age of 1 year, it is possible that these signs are markers of later cognitive and learning disorders as have been described previously.⁸ The planned follow-up at the age of 7 years will include an assessment of both cognitive function and school performance, together with testing for right hemisphere deficit syndrome.¹²

The infants were very ill when recruited into the trial. It is not known if the morbidity among survivors would have been altered if ECMO had been offered at an earlier stage, that is before the oxygenation index reached 40. It is also possible that other therapies such as inhaled nitric oxide and oscillation ventilation introduced early might alter outcome or the need for ECMO. Such approaches to treatment are yet to be fully evaluated.

The extent to which respiratory or neurologic morbidity among these very sick infants reflects the underlying condition or the treatment offered has been a source of debate.^{13,14} Comparisons of outcome between nonrandomized groups of infants are difficult to interpret.^{15,16} In this trial, however, randomized groups were comparable at trial entry, and it is possible to gain some unique insights into the relationship of preexisting condition, treatment, and outcome. Both modes of treatment carry risks as well as benefits. Although there were more survivors with disability not classified as severe in the ECMO group, this appeared to reflect the increased survival in this group, rather than any specific adverse effect of ECMO. A summary of these results has been sent to parents who wished to receive them. A qualitative study is ascertaining their views about these and other aspects of the trial.¹⁷

With the exception of diaphragmatic hernia where doubt still remains, the clinical effectiveness of ECMO for neonates who meet the trial criteria is now proven. The use of ECMO outside this group will be a matter of judgement. ECMO use among older children and adults is increasing, and there have been no

rigorous evaluations of the type reported here among these sorts of patients. In respect of neonates, there is an argument for transporting infants to an ECMO center before their condition becomes critical. It is not clear, however, whether the extra costs in the short-term would be offset by later benefits.

The results of the economic evaluation, carried out alongside the trial, based on the principal endpoint of death or severe disability at age 1, found the additional cost of ECMO per additional survivor to be within the range of other life-extending technologies such as renal transplantation¹⁸. The potential long-term economic implications for the health, educational, and social services and the families are being addressed alongside the 4- and 7-year follow-up studies. The economic justification for introducing ECMO in any particular setting, however, depends not only on effectiveness but also issues of access and likely workload. These issues will be discussed in more detail elsewhere.

Because ECMO was introduced into the UK relatively late in comparison to North America, this situation did provide an opportunity to evaluate a new technology when it was relatively mature. Although there was uncertainty among British pediatricians about its value, there was voluntary agreement to limit its use amongst neonates across the UK solely to the trial. This undoubtedly contributed significantly to the success of the trial and allowed evaluation of ECMO, both in terms of its clinical effectiveness and its cost effectiveness, before it became widely adopted. Although controversial, we think it is a model that should be considered for the evaluation of new health technologies more generally.

UK Collaborative ECMO Trial Group

Writing Committee

A. Johnson, D. Field, D. Elbourne, and A. Grant

Steering Group

D. Field (Chairman), C. Davis, D. Elbourne (Trial Coordinator), A. Grant (Trial Coordinator), A. Greenough, P. Hale, L. Hamilton, A. Johnson, M. Levene, M. Liddell, F. Lockett, D. Macrae, and C. Skeoch

Advisory Groups

Follow-up: D. Elbourne, D. Field, A. Grant, A. Johnson, M. Levene, and R. Morley. *Health Economics:* D. Elbourne, D. Field, A. Grant, A. Johnson, L. Hallam, S. Howard, M. Mugford, C. Normand, and T. Roberts. *Pathology:* J. Bell, D. Doyle, D. Elbourne, M. Evans, D. Field, A. Grant, A. Howatson, I. Jeffrey, P. McKeever, R. Risdon, W. Squier, and C. Wright. *Respiratory:* C. Beardsmore, D. Elbourne, D. Field, A. Johnson, and J. Stocks. *Technical:* C. Davis, D. Elbourne, M. Elliott, D. Field, R. Firmin, and L. Hamilton

Data Monitoring Committee

R. Doll (Chairman), E. Alberman, D. Altman, F. Cockburn, and R. Cooke

ECMO Trial Clinical Coordinating Center

D. Field, A. Fenton, J. Grant, W. Hoskyns, U. MacFadyen, A. Doulah (1995), S. Kerr (1994), and G. Pearson (1993)

Developmental Pediatricians

M. Graham, C. Jessen, and A. Schreuder

H. Ashurst (Neonatal Perinatal Epidemiology Unit Computing Coordinator), S. Ayers (Programmer), C. Bridgwood (Administrative Coordinator), D. Elbourne (Director, Perinatal Trials Service), K. Enock (Administrative Coordinator), J. Fooks (Programmer), C. Harris (Data Manager), A. Johnson (Pediatrician), E. Lukes (Data Entry), and A. Wrotchford (Programmer)

ECMO Centers: (local medical coordinators, ECMO fellows, nursing coordinators, pediatric surgeons, and pathologists):

Freeman Newcastle (J. Hamilton, J. Madar, G. Derrick, L. Scott, and C. Wright); Glenfield General Leicester (G. Pearson, S. Kerr, A. Doulah, H. Moore, and R. Firmin); Great Ormond Street (D. Macrae, C. Pierce, L. Tyszczyk, A. Goldman, B. Boosfeld, E. Smith, and R. Risdon); King's College (A. Greenough, V. Noble, M. King, J. Desai, and M. Newbould); and Royal Hospital for Sick Children Glasgow (C. Skeoch, A. Tometzki, E. Thomson, M. Liddell, C. Davis, and A. Howatson)

Referral Hospitals: (local medical coordinators, nursing coordinators, pediatric surgeons, and pathologists)

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APPENDIX. Details of Survivors With Impairment and/or Disability at Age 1 Year

Allocation to ECMO	Description of Outcome	Diagnosis at Trial Entry
Severe disability (1)	Spastic quadriplegia, low vision, severe developmental delay. Albinism.	Meconium aspiration
Impairment with functional loss (12)	Left hemiplegia	Meconium aspiration
	Motor delay (DQ = 54). Variable tone on left.	Idiopathic respiratory distress syndrome
	Truncal hypertonia. Strong left-hand preference.	Persistent fetal circulation
	Additional oxygen until 11 months age, tube feeding, tone change on left. In hospital at 1 year.	Diaphragmatic hernia
	Additional oxygen at night.	Persistent fetal circulation
	Sensorineural deafness—bilateral hearing aids.	Sepsis
	Hypopituitarism, global delay, oxygen dependent, tube feeding.	Meconium aspiration
	Transposition of great vessels. Motor delay (DQ = 57).	Meconium aspiration
	Right hemiplegia.	Meconium aspiration
	Hypotonic, requiring tube feeding.	Diaphragmatic hernia
	Mild left hemiplegia, some motor delay.	Persistent fetal circulation
	Hypotonia—motor delay.	Diaphragmatic hernia
Impairment without functional loss (4)	Mild asymmetry of hand and arm use—left sided signs.	Sepsis
	Mild asymmetry of hand use—left sided signs. On anticonvulsants in first month.	Meconium aspiration
	Left hand preference.	Idiopathic respiratory distress syndrome
	Increased tone in left leg.	Idiopathic respiratory distress syndrome
Allocation to Conventional Care	Description of Outcome	Diagnosis at Trial Entry
Severe disability (1)	Severe global delay. Maternal anticonvulsants in pregnancy. (Overall DQ = 47)	Idiopathic respiratory distress syndrome
Impairment with functional loss (4)	Motor delay (Motor DQ = 64), abnormal tone patterns.	Persistent fetal circulation
	Severe chronic lung disease, oxygen dependent, gastrostomy. In hospital at 1 year. Severe motor delay (Motor DQ = 29).	Idiopathic respiratory distress syndrome
	Needing oxygen at night.	Idiopathic respiratory distress syndrome
	Seizures, increased tone on left.	Meconium aspiration
Impairment without functional loss (5)	Ventriculoperitoneal shunt for hydrocephalus. Normal development.	Persistent fetal circulation
	Reduced tone—left leg.	Persistent fetal circulation
	Mild asymmetry—left-sided signs.	Idiopathic respiratory distress syndrome
	Poor attention, possible hearing loss. Mild global delay.	Meconium aspiration
	Hypotonic lower limbs.	Persistent fetal circulation

REFERENCES

1. UK Collaborative ECMO Trial Group. UK collaborative randomised trial of neonatal extracorporeal membrane oxygenation. *Lancet*. 1996; 348:75–82
2. Griffiths R. *The Abilities of Young Babies*. High Wycombe, England: Association for Research in Infant and Child Development; 1970
3. Dundas I, Beardsmore CS, Enock K, Poole K, Stocks J. Respiratory function at 1 year in survivors of UK ECMO Trial. *Am J Respir Crit Care Med*. 1997;155:A237
4. Howard S, Normand C, Mugford M, et al. Costing neonatal care alongside the collaborative ECMO trial: how much primary research is required? *Health Econ*. 1995;4:265–271
5. Gross SJ, Bifano EM, Gugenio D, Hakanson DO, Hingre RV. Prospective randomized controlled trial of conventional treatment or transport for ECMO in infants with severe persistent pulmonary hypertension (PPHN): two year follow up. *Pediatr Res*. 1994;36:17A
6. Baumgart S, Streletz LJ, Needleman L, et al. Right common carotid artery reconstruction after extracorporeal membrane oxygenation: vascular imaging, cerebral circulation, electroencephalographic, and neurodevelopmental correlates to recovery. *J Pediatr*. 1994;25:295–304
7. Schumacher RE, Palmer TW, Roloff DW, LaClaire PA, Bartlett RH. Follow-up of infants treated with extracorporeal membrane oxygenation for newborn respiratory failure. *Pediatrics*. 1991;87:451–457
8. Glass P, Wagner AE, Papero PH, et al. Neurodevelopmental status at age five years of neonates treated with extracorporeal membrane oxygenation. *J Pediatr*. 1995;27:447–457
9. Mendoza JC, Shearer LL, Cook LN. Lateralization of brain lesions following extracorporeal membrane oxygenation. *Pediatrics*. 1991;88: 1004–1009
10. Hendricks-Mendoza KD, Walton J. Hearing loss in infants with persistent fetal circulation. *Pediatrics*. 1988;81:650–656
11. Kawashiro N, Tsuchihashi N, Koga K, Kawano T, Itoh Y. Delayed post-neonatal intensive care unit hearing disturbance. *Int J Pediatr Otorhinolaryngol*. 1996;34:35–43
12. Voeller KKS. Right hemisphere deficit syndrome in children. *Am J Psychiatry*. 1986;143:1004–1009
13. Bernbaum J, Schwartz IP, Gerdes M, D'Agostino JA, Coburn CE, Polin RA. Survivors of extracorporeal membrane oxygenation at 1 year of age:

- the relationship of primary diagnosis with health and neurodevelopmental sequelae. *Pediatrics*. 1995;96:907–913
14. Robertson CM, Finer NN, Sauve RS, et al. Neurodevelopmental outcome after neonatal extracorporeal membrane oxygenation. *Can Med Assoc J*. 1995;152:1981–1988
 15. Walsh-Sukys MC, Bauer RE, Cornell DJ, Friedman HG, Stork EK, Hack M. Severe respiratory failure in neonates: mortality and morbidity rates and neurodevelopmental outcome. *J Pediatr*. 1994;125:104–110
 16. Wildin SR, Landry SH, Zwischenberger JB. Prospective controlled study of developmental outcome in survivors of extracorporeal membrane oxygenation: the first 24 months. *Pediatrics*. 1994;93:404–408
 17. Snowdon C, Garcia J, Elbourne D. Making sense of randomization: responses of parents of critically ill newborn babies to random allocation of treatment in a clinical trial. *Soc Sci Med*. 1997;45:1337–1355
 18. Roberts T on behalf of ECMO Economics Group. Economic evaluation alongside the UK Collaborative ECMO Trial. Proceedings of the Royal College of Pediatrics and Child Health Annual Meeting; 1997. 1;61

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